

Global Pandemic Case Study Rubric

DS 4002 – Spring 2026 – Blake Richardson

Submission format: Upload link to GitHub repository on UVA Canvas

Individual Assignment

Why am I doing this?

This case study allows you to apply your data science knowledge to analyze and suggest real solutions to public health crises. By analyzing COVID-19 pandemic data across different continents, you will gain hands-on experience in data processing, synthesis, analysis, and visualization using time-series modeling.

What am I going to do?

The GitHub repository for this case study can be found at https://github.com/BlakeSRichardson/DS4002_CS3. You will obtain the referenced dataset from 'Our World in Data,' cleaning and processing the data for model use. After reviewing provided information on SARIMA, you will fit SARIMA to each of 6 continents to capture regional long-term and seasonal trends. Using research background provided on the COVID-19 pandemic to select a reasonable cutoff date, you will train the model on data prior to *your chosen cutoff date* and test the model on data after *your chosen cutoff date* to determine how well different continents handled later stages of the pandemic. Using the R-squared test and data visualizations, you will evaluate how accurate the model performed in comparison with actual outcomes. Additionally, you will use the chi-squared test of independence to determine whether continent outcomes varied in a statistically significant way. Finally, apply your findings and knowledge of the COVID-19 pandemic to suggest successful strategies for future pandemics.

Your final deliverables should include:

- A data dictionary
- Well-detailed source code with labeled comments
- Data Visualizations of each Continent's Actual v. Predicted Outcomes
- A Document of Potential Pandemic Intervention Strategies
- A GitHub repository containing all materials used

Tips for success:

- Review provided documentation on how to successfully use SARIMA models
- Familiarize yourself with Python (time series modeling and visualizations) and R (statistical testing)
 - For statistical testing: review provided instructions on how to properly produce and interpret chi-squared test of independence and R-squared test outputs
- Be deliberate in selecting a cutoff date that corresponds to an important event in the COVID-19 pandemic
- Clearly label selected columns of data (infection rate/death rate) and remove additional non-continent categories (by country)

How will I know I have succeeded?

You will meet expectations on this case study when you successfully follow and complete the criteria in the rubric below:

Spec Category	Spec Details
Formatting	• One GitHub repository (submitted via link on Canvas) □ Create a new GitHub repository for this assignment titled 'CS2_GlobalPandemic' that contains □ README.md □ LICENSE.md □ Source Code File □ List of Proposed Pandemic Interventions/Strategies □ Your data output (i.e., visualizations of each continent's outcomes) □ REFERENCES.md
README.md	• Brief summary of what files your Github repository contains and their relevance to project objectives

Source Code File	Clearly labeled (through comments) Jupyter Notebook file and R Script that contains code used to clean/preprocess data, train/test SARIMA, produce data visualizations, and produce statistical analysis output. In the source code, you must include: • 6 Visualizations of each Continent's Predicted v. Actual Outcomes • Commented Explanation of each Graph and its Implications • R-Squared Test Table & Results • Chi-Square Test Table & Results
REFERENCES.md	Include a markdown file in repository that lists all references used in IEEE Documentation Style • Under each source, explain (1-3 sentences each) how the document assisted you in your analysis

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